

BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Midterm

Duration: 1 Hour 10 Minutes

Set - A

Semester: Fall 2024

Full Marks: 30

CSE 420: Compiler Design

Figures in the right margin indicate marks.

Answer all the questions

<u>COs</u>	<u>Questions</u>	<u>Marks</u>
CO1	1. Draw the architecture diagram of a typical compiler showing the different steps. Write one/two sentences explaining what each step does.	2+3=5
CO3	2. Draw the LR(0) automaton for the following grammar. <i>(You will lose 0.5 point for each missing state, wrong LR(0) items in a state, and missing/incorrect transition arrows.)</i> A -> N O E N -> id O -> = O -> '+=' O -> '-=' E -> E + N E -> N Here the alphabet of terminal symbols is {=, '+=', '-=', +, id}	10
CO3	3. Compute the first and follows of non-terminal symbols for the following augmented grammar. A' -> A A -> BC B -> D C -> N*D C -> ε N -> id D -> num D -> ε The alphabet of terminal symbols is {*, id, num}. Don't forget to consider the end-marker \$.	2+3=5

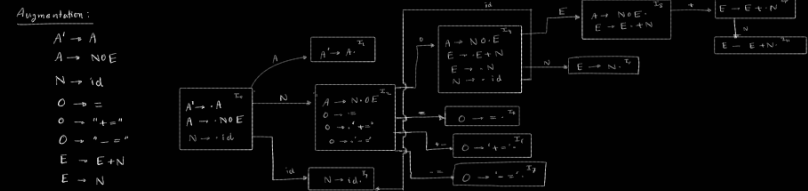
4. Consider the following grammar and look at the SLR(1) parse table below:

1. $E \rightarrow E + T$
2. $E \rightarrow T$
3. $T \rightarrow T * F$
4. $T \rightarrow F$
5. $F \rightarrow (E)$
6. $F \rightarrow id$

STATE	ACTION						GOTO		
	id	+	*	(	)	\$	E	T	F
0	s5			s4			1	2	3
1		s6				acc			
2		r2	s7		r2	r2			
3		r4	r4		r4	r4			
4	s5			s4			8	2	3
5		r6	r6		r6	r6			
6	s5			s4				9	3
7	s5			s4					10
8		s6			s11				
9		r1	s7		r1	r1			
10		r3	r3		r3	r3			
11		r5	r5		r5	r5			

Show how an SLR(1) parser with following grammar rules and parsing tables process the following input string. id+id*id

Answer to the question no 02



Answer to the question no 03

$A' \rightarrow A$
 $A \rightarrow B \mid C$
 $C \rightarrow N * D$
 $C \rightarrow E$
 $N \rightarrow id$
 $D \rightarrow num$
 $D \rightarrow E$

First (A) = {num, id, E}
First (B) = {num, id, E}
First (C) = {id, E}
First (D) = {num, E}
First (N) = {id}

Follow (A') = {A}
Follow (A) = {A}
Follow (B) = {id, A}
Follow (C) = {A}
Follow (D) = {id, A}
Follow (N) = {A}

$C \rightarrow N * D$
 $A \rightarrow B \mid C$
 $B \rightarrow id$
 $E \rightarrow N$
 $A \rightarrow B \mid C$
 $B \rightarrow id$
 $E \rightarrow N$

Answer to the question no 4

Stack	Symbol	Input Buffer	Action
0	\$	id * id * id \$	S6
05	* id	* id * id \$	Reduce by $F \rightarrow id$
03	\$ F	* id * id \$	Reduce by $T \rightarrow F$
02	\$ T	* id * id \$	Reduce by $E \rightarrow T$
01	* E	* id * id \$	S6
016	\$ E *	* id * id \$	S6
015	\$ E * id	* id \$	Reduce by $F \rightarrow id$
013	* E * F	* id \$	Reduce by $T \rightarrow F$
012	* E * T	* id \$	S7
010	* E * T *	id \$	S5
009	\$ E * T * id	\$	Reduce by $F \rightarrow id$
01122*id*	* E * T * F	\$	Reduce by $T \rightarrow T * F$
0102	* E * T	\$	Reduce by $E \rightarrow E * T$
01	\$ E	\$	Accept